PRADEEP PUJARI

Research Engineer — Formal Verification, Theorem Proving & AI Systems

510-894-1468 | ppujari@hotmail.com | LinkedIn | GitHub | Kaggle | Google Scholar | Remote / SF Bay Area

SUMMARY

Software engineer with 20+ years building production search, NLP and LLM systems, now focused on **formally verified software and AI-assisted theorem proving**. Building **RegLib**, a formally verified compliance engine (**Lean 4, Z3, first-order and deontic logic**), and post-training Gemma 4 into a **SWE coding agent** for Kaggle. Background in compilers (**Unix lex/yacc**, LLVM-IR), developer tooling, Kubernetes-based deployment and reproducible ML pipelines. Open-source contributor to Apache Lucene-Solr and EleutherAI.

SELECTED PROJECTS — FORMAL VERIFICATION & AI AGENTS

RegLib — Formally Verified Compliance Engine (Lean 4, FOL) *Independent research*

- Architecting a pipeline that turns capital-markets regulation (**SEBI ICDR 2018**) into machine-checkable rules: structural extraction, RAFT-inspired LLM **auto-formalization** into a **first-order-logic intermediate representation**, and **deontic norm** formalization (obligation / permission / prohibition).
- Encoding rules and building proofs in the **Lean 4** proof assistant, with **Z3 (SMT)** as the compliance-checking engine, so a verdict on an IPO filing rests on a checked proof or counterexample rather than on untrusted LLM output.
- Six-phase plan covering extraction, knowledge explication, norm formalization, Lean proof construction, the Z3 engine, and GamePad-inspired hierarchical embeddings for proof-state representation.

Kaggle Gemma 4 Developer Agent — SWE coding agent *In progress*

- Post-training **Gemma 4 31B (QAT, w4a16)** with **LoRA adapters** (under 3 GiB) into an agent that navigates complex codebases and drafts fixes for real software issues on everyday hardware; scored by **resolution rate** on SWE-style tasks.
- Building the declarative agent configuration (agent.yaml) and evaluation config, and tuning tool-use behavior to fit per-task budgets (tool calls and wall-clock time).

Compiler built with Unix lex/yacc *Systems programming*

- Designed the grammar and built a parser/compiler with **Unix lex/yacc**: the same foundations (parsers, analyzers, transpilers) that verification tooling is built on. Later applied **LLVM-IR** analysis and optimization at Meta.

EleutherAI — Alignment Research (Open Source) *SOAR*

- Researching LLM alignment for legal reasoning by integrating the **Lean theorem prover** with agentic retrieval-augmented systems to formalize and verify compliance rules, improving transparency, traceability and constraint adherence.

TECHNICAL SKILLS

Formal methods & verification: Lean 4 (proof assistant) / SMT solving, first-order logic, deontic logic, autoformalization, LLM-assisted proof and spec generation

Languages: Python, Java, C++, SQL, C (lex/yacc); LLVM-IR

Dev tools & compilers: Parsers, grammars, code analysis, LLVM-IR, Apache TVM, ONNX, inference optimization (quantization, GPTQ/AWQ)

CI/DevOps & infrastructure: Kubernetes, Azure, AWS SageMaker, Apache Ray, Feast feature store, Kafka, Spark/PySpark, reproducible experiment pipelines, UNIX internals

AI systems & agents: PyTorch, Hugging Face, LoRA/PEFT, RLHF, DPO, GRPO, PPO; LangChain, LangGraph, AutoGen, ReAct; RAG / GraphRAG; LLM-as-judge evaluation

Search & data: Lucene, Solr, Elasticsearch, FAISS, ChromaDB, Cassandra, MongoDB, PostgreSQL, Redis

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer (Contract) — Dick's Sporting Goods May 2025 – Jul 2026

- Designed automated evaluation loops with LLM-as-judge (Claude, LLaMA) tracking NDCG and Recall@k, enabling continuous tuning of retrieval quality without manual labeling at scale
- Designed and productionized vector search pipelines with fine-tuned text embedding models for high-recall retrieval and multi-stage reranking for precision optimizing product discovery for millions of items

Technologies: Python, Azure Foundry, E5, SBERT, DeBERTa, LLM-as-Judge, ChromaDB, PySpark

GenAI Architect – KP Digital: Jul 2022-Apr 2025

- Designed and scaled post-training pipelines to align open-weight LLMs (LLaMA, MedAlpaca, T5) to domain-specific tasks using SFT, RLHF, DPO, GRPO, and RLAIF-resulted reduction in unsafe responses, improved instruction-following on domain tasks

- Designed and implemented a multi-agentic conversational AI chatbot for customer service using a RAG architecture to reduce hallucination, cutting support call volume by 5% while improving response accuracy.
- Developed NLP Agentic pipelines for literature mining to extract insights from scientific texts and biomedical databases; enabled PII anonymization, hypothesis generation, and knowledge curation.

Technologies: Python, PyTorch, Azure Foundry, Kubernetes, vLLM, LangChain, LangGraph, LlaMA, MCP

Software Engineer - Meta: Dec 2021-Jul 2022

- Benchmarked and optimized **data and model parallelism** for a large-scale **SparseNN Ad Ranking ML model** on cutting-edge AI accelerators and chips, improving performance and efficiency for Meta's high-traffic advertising infrastructure.
- Developed and enhanced a **personalization model** that tailored ad content to individual users preferences, leveraging real-time data and optimizing for both relevance and engagement
- Applied QLoRA for memory-efficient fine-tuning, and post-training quantization with hyperparameter tuning to accelerate inference while maintaining model accuracy.
- Leveraged LLVM-IR to perform advanced code analysis and optimizations, enabling efficient translation of high-level languages into optimized machine code for multiple architectures.

Technologies : Python, PyTorch, Quantization Techniques, LLM, Python, FSDP, DDP

Principal Research Scientist (ML) - Kohl's Innovation Lab Aug 2017-May 2020

- Implemented real-time object detection, segmentation, gesture recognition system non-rigid tracking with pre-trained deep learning models YOLO, SSD
- Store Shopping Intent Analysis models for people counter, Age and Gender identification.
- Optimized and deployed ML models on edge devices using Apache TVM for efficient inference.
- Built an ML model that predicts item returns at the point of sale from historical return data, helping reduce costly return processing, reticketing, and restocking.

Technologies: Python, PyTorch, Azure Foundry, ONNX, DNN, CLIP, XGBoost, Scikit-Learn

Staff Machine Learning Engineer - Walmart Labs May 2010-Mar 2017

- Built ML models for attribute extraction such as detecting brand, color, size etc. with
  - Supervised Learning B. Sequence Labelling and CRF C. **CNN** approach.
- Implemented Neural Network model for Item Matching System based on title and description.
  - Used pretrained Word embedding – **Glove**-contextual word representation.
- Worked in distributed data pipeline to orchestrate raw item JSON through a series of **micro services** producing a sellable item.
- High-Performance Computing (HPC) workloads: Built distributed machine learning frameworks to train deep neural networks on massive datasets.

Technologies: Python, Scikit Learn, **TensorFlow**, **Keras**, ZooKeeper, Kafka, **Storm**, Cassandra, Mongo

Earlier Career: Team Lead — Search Science & ML | ServiceNow • CVS Health • Angie's List • Macy's.com | 2001–2014

- Implemented offline automatic IR Evaluation, Real- Time Indexing, Search Ranking Algorithm
- Implemented Location based Search, Indexing Pipeline, Vector Search, Nutch-based crawler, NER

Technology: Java J2EE, Tomcat, SpringBoot, ETL, Data Lakes, Lucene-Solr, Elastic Search, Learn to Rank, Knowledge Graph, MS-Marco, Data Modelling, RDBMS, NoSQL

EDUCATION, PUBLICATIONS & LEARNING

- MS in Computer Science and Applications - National Institute of Technology, India (June 1988)
- AI Specialization - Stanford University (Jul 2023)
- Detecting Cyberbullying Instances, published paper (Apr 2023)
- Coauthor, Practical Convolutional Neural Networks, Packt Publishing (Feb 2018)